Lab03 Spring 2026 Section

2026-02-15

(Textbook 2.23) Refer to Copier maintenance Problem 1.19

a. Set up the ANOVA table.

```
df19 = read.table("CH01PR19.txt",
                  header=FALSE, sep="",
                  col.names=c("y","x"))

lmFit19 = lm(y~x, data = df19)
anova(lmFit19)
```

```
## Analysis of Variance Table
##
## Response: y
##            Df Sum Sq Mean Sq F value   Pr(>F)
## x           1  3.588  3.5878  9.2402 0.002917 **
## Residuals 118 45.818  0.3883
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

b. What is estimated by MSR in your ANOVA table? by MSE? Under what condition do MSR and MSE estimate the same quantity?

   You need to look at your ANOVA table for the values. MSR = Mean SQ of X = 3.5878 MSE = Mean SQ of residulas = .3883 They would estimate the same quantity when b1=0. Your slope has to be 0 for MSE = MSR.

c. Conduct an F test of whether or not B1 = O. Control the a risk at .01. State the alternatives, decision rule, and conclusion.

```
f = 3.5878 / .3883
print(f)
```

```
## [1] 9.239763
```

alternatives & decision rule: H_0: B_1 =0, H_a:B_1 !=0

Conclusion: reject the null since p-value is less than our alpha (risk) at .01.

P-value: (found in anova): .002917 < .01

With the question in the homework, you have to take it further because you have a train and test dataset now

Step #1 run the regression problem above and subtract some objects to use later. This is creating your coefficient objects

Step #2 obtain the F statistics and p-value associated with the regression

Step #3 apply the decision rule if p-value > 0.01 (this is from this example - what does your homework say?)

Step #4 get the F critical

Step #5 Confirm numerical equivalence - #Is the P-value for the F test the same as that for the t test?

d. What is the absolute magnitude of the reduction in the variation of Y when X is introduced into the regression model? What is the relative reduction? What is the name of the latter measure?

```
#you can find SSE and SSR from lmFit19
anova(lmFit19)
```

```
## Analysis of Variance Table
##
## Response: y
##            Df Sum Sq Mean Sq F value   Pr(>F)
## x           1  3.588  3.5878  9.2402 0.002917 **
## Residuals 118 45.818  0.3883
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
SSE = 45.818
SSR = 3.588

#you can find SSE(R) from lmFit19_r when x= 1 (this is our reduced regression)
lmFit19_r = lm(formula = y ~ 1, data= df19)
anova(lmFit19_r)
```

```
## Analysis of Variance Table
##
## Response: y
##            Df Sum Sq Mean Sq F value Pr(>F)
## Residuals 119 49.405 0.41517
```

```
# SSE(R) = 49.405

#rr = SSR / SSE
rr = 3.588/49.405
print(rr)
```

```
## [1] 0.07262423
```

Relative reduction: 0.07262423

(Textbook 2.30) Refer to Crime rate Problem 1.28.

a. Test whether or not there is a linear association between crime rate and percentage of high school graduates, using a t test with a = .01. State the alternatives, decision rule, and conclusion. What is the P-value of the test?

```
df_crimeRate = read.table("CH01PR28.txt",
                          header=FALSE, sep="",
                          col.names=c("cr","ed"))
lmFit1_28 = lm(cr~ed, data=df_crimeRate)
summary(lmFit1_28)
```

```
##
## Call:
## lm(formula = cr ~ ed, data = df_crimeRate)
##
```

```
## Residuals:
##     Min      1Q  Median      3Q     Max 
## -5278.3 -1757.5  -210.5  1575.3  6803.3 
## 
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)    
## (Intercept) 20517.60    3277.64   6.260 1.67e-08 ***
## ed           -170.58      41.57  -4.103 9.57e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## 
## Residual standard error: 2356 on 82 degrees of freedom
## Multiple R-squared:  0.1703, Adjusted R-squared:  0.1602 
## F-statistic: 16.83 on 1 and 82 DF,  p-value: 9.571e-05
```

```r
#here we are working with a t-sided t-test

#calculcate t-value
t = -170.58 / 41.57
print(t)
```

```
## [1] -4.10344
```

```r
#calculate critical value
#our degrees of freedom is showing that you have about 84 observations minus 2 (estimating 2 unknown pa
critical = qt(0.995, df = 82)
print(critical)
```

```
## [1] 2.637123
```

alternatives: H_0: B_1= 0 H_a: B_1 neq 0

t-value: -4.103 p-value (found in summary): 9.571e-05

decision rule: reject H_0 if |t| > 2.637 or reject H_0 if the p-value <.01

conclusion: So we reject the null and say that our regression model is significant